W2-PM4: theHarvester-Based Footprinting

Kali Linux | theHarvester 4.10.1 | Passive OSINT

1. Objective

To use the theHarvester OSINT tool to gather publicly available information about a domain and identify its digital footprint.

2. Tools Used

Tool	Purpose
theHarvester 4.10.1	OSINT / footprinting
Kali Linux	Operating environment
Certificate Transparency (crt.sh)	Data source

3. Target Domain

tesla.com

Only passive, publicly available sources were queried. No active scanning of the target's systems was performed.

4. Methodology

Step 1 - Verify the tool

```
theHarvester -h
```

Step 2 - Run the footprinting query

```
theHarvester -d tesla.com -b crtsh
```

The -d option specified the target domain, while -b crtsh selected Certificate Transparency as the information source.

Step 3 - Save the results (JSON and XML)

```
theHarvester -d tesla.com -b crtsh -f tesla_footprinting
```

Step 4 - Verify output files

```
ls -lh tesla_footprinting*
```

Step 5 - Inspect the JSON results

```
cat tesla_footprinting.json
```

Step 6 - Calculate statistics

```
python3 -c 'import json; d=json.load(open("tesla_footprinting.json"));
print("Total discovered hosts:",len(d.get("hosts",[])));
print("Email addresses:",len(d.get("emails",[])));
print("Shodan results:",len(d.get("shodan",[])))'
```

5. Results

Finding	Result
Discovered hostnames	236

Email addresses	0
Shodan results	0

Examples of discovered hostnames:

- sso.tesla.com
- energy.tesla.com
- fleet-auth.tesla.com
- inference.tesla.com

Output file	Size
tesla_footprinting.json	6.9K
tesla_footprinting.xml	9.2K

6. Security Analysis

The results demonstrate how publicly available certificate records can reveal hostnames and provide insight into an organisation's digital footprint.

Such information can help security teams understand their external attack surface and identify assets requiring inventory and monitoring.

Important: the discovery of a hostname does not establish that the host is active or vulnerable.

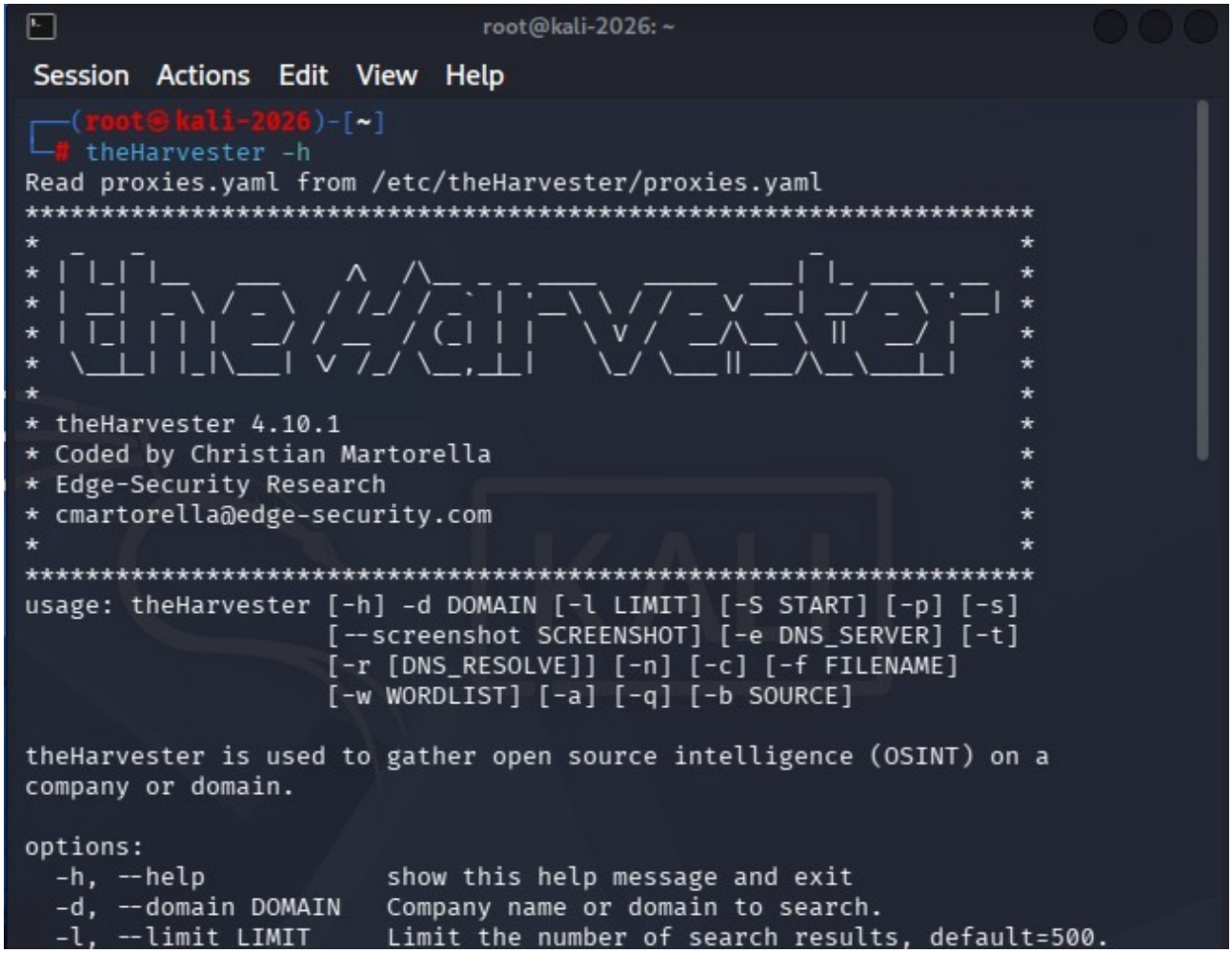
7. Conclusion

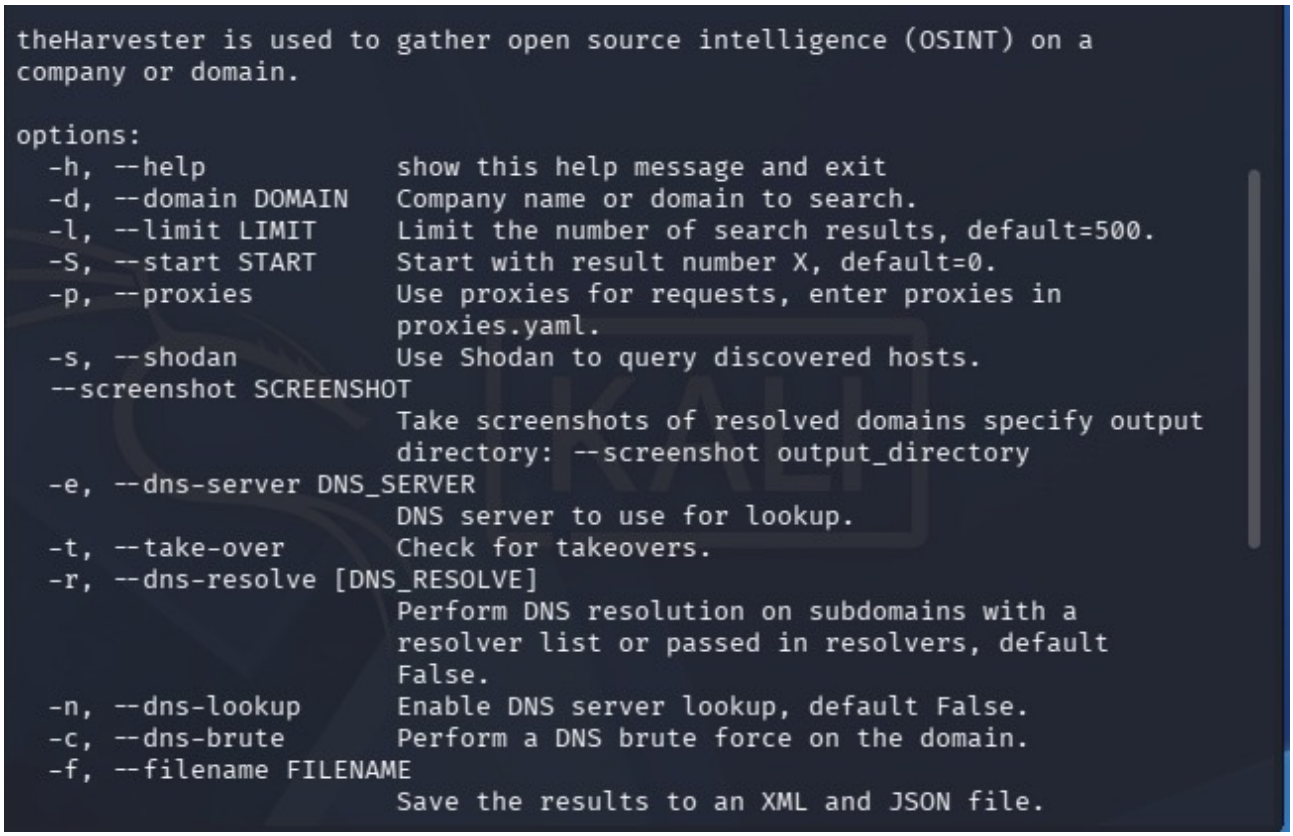
The practical successfully demonstrated passive footprinting using theHarvester. A total of 236 hostname entries were collected, with no email addresses or Shodan results returned.

The findings were exported into JSON and XML formats for further analysis and documentation.

8. Evidence

Screenshot 1: Tool verification


root@kali-2026: ~
Session Actions Edit View Help
┌──(root㉿kali-2026)-[~]
└─# theHarvester -h
Read proxies.yaml from /etc/theHarvester/proxies.yaml
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-S START] [-p] [-s]
[--screenshot SCREENSHOT] [-e DNS_SERVER] [-t]
[-r [DNS_RESOLVE]] [-n] [-c] [-f FILENAME]
[-w WORDLIST] [-a] [-q] [-b SOURCE]
theHarvester is used to gather open source intelligence (OSINT) on a
company or domain.
options:
-h, --help show this help message and exit
-d, --domain DOMAIN Company name or domain to search.
-l, --limit LIMIT Limit the number of search results, default=500.


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company or domain.
options:
-h, --help show this help message and exit
-d, --domain DOMAIN Company name or domain to search.
-l, --limit LIMIT Limit the number of search results, default=500.
-S, --start START Start with result number X, default=0.
-p, --proxies Use proxies for requests, enter proxies in
proxies.yaml.
-s, --shodan Use Shodan to query discovered hosts.
--screenshot SCREENSHOT
Take screenshots of resolved domains specify output
directory: --screenshot output_directory
-e, --dns-server DNS_SERVER
DNS server to use for lookup.
-t, --take-over Check for takeovers.
-r, --dns-resolve [DNS_RESOLVE]
Perform DNS resolution on subdomains with a
resolver list or passed in resolvers, default
False.
-n, --dns-lookup Enable DNS server lookup, default False.
-c, --dns-brute Perform a DNS brute force on the domain.
-f, --filename FILENAME
Save the results to an XML and JSON file.

```
  -r, --dns-resolve [DNS_RESOLVE]
                        Perform DNS resolution on subdomains with a
                        resolver list or passed in resolvers, default
                        False.
  -n, --dns-lookup      Enable DNS server lookup, default False.
  -c, --dns-brute       Perform a DNS brute force on the domain.
  -f, --filename FILENAME
                        Save the results to an XML and JSON file.
  -w, --wordlist WORDLIST
                        Specify a wordlist for API endpoint scanning.
  -a, --api-scan        Scan for API endpoints.
  -q, --quiet           Suppress missing API key warnings and reading the
                        api-keys file.
  -b, --source SOURCE   baidu, bevigil, bitbucket, brave, bufferoverun,
                        builtwith, censys, certspotter, chaos, commoncrawl,
                        criminalip, crtsh, dehashed, dnsdumpster,
                        duckduckgo, fofa, fullhunt, github-code, gitlab,
                        hackertarget, haveibeenpwned, hudsonrock, hunter,
                        hunterhow, intelx, leakix, leaklookup, netlas,
                        onyphe, otx, pentesttools, projectdiscovery,
                        rapiddns, robtex, rocketreach, securityscorecard,
                        securityTrails, shodan, subdomaincenter,
                        subdomainfinderc99, thc, threatcrowd, tomba,
                        urlscan, venacus, virustotal, waybackarchive,
                        whoisxml, windvane, yahoo, zoomeye

┌──(root㉿kali-2026)-[~]
└─#
```

Screenshot 2: Footprinting execution

```
*.cn.tesla.com
*.de.tesla.com
*.eu.logs.tesla.com
*.gf.tesla.com
*.logs.tesla.com
*.na.logs.tesla.com
*.nv.tesla.com
*.ny.tesla.com
*.paloalto.tesla.com
*.powerhub.energy.tesla.com
*.tx.tesla.com
CitiApiPgpProdV8.tesla.com
CitiApiSslProdV8.tesla.com
akamai-apigateway-einvoicing-stg.tesla.com
akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com
akamai-apigateway-vehicleextinfogw-stgsvc-st.tesla.com
ams13-gpgw1.tesla.com
apac-cppm.tesla.com
apac.logs.tesla.com
apf-api.eng.vn.cloud.tesla.com
apf-api.prd.vn.cloud.tesla.com
api-firebolt.tesla.com
assets.github-ap.tesla.com
```

Screenshot 3: Saved scan results

```
ui.a.smf11.tcs.tesla.com
ui.b.smf11.tcs.tesla.com
ui.c.smf11.tcs.tesla.com
ui.d.smf11.tcs.tesla.com
uploads.github-ap.tesla.com
uploads.github-fw.tesla.com
uploads.github-it.tesla.com
uploads.github.tesla.com
vdi.tesla.com
vehicle-files.eng.euw1.vn.cloud.tesla.com
vehicle-files.eng.usw2.vn.cloud.tesla.com
vehicle-files.prd.euw1.vn.cloud.tesla.com
vehicle-files.prd.usw2.vn.cloud.tesla.com
viewscreen.github-ap.tesla.com
viewscreen.github-fw.tesla.com
viewscreen.github-it.tesla.com
viewscreen.github.tesla.com
vrp-stg.tesla.com
x3.obs.tesla.com

[*] Reporting started.
[*] XML File saved.
[*] JSON File saved.


┌──(root㉿kali-2026)-[~]
└─#
```

Screenshot 4: Output file verification

```
┌──(root㉿kali-2026)-[~]
└─# ls -lh tesla_footprinting*
-rw-rw-r-- 1 root root 6.9K Nov 11 11:50 tesla_footprinting.json
-rw-rw-r-- 1 root root 9.2K Nov 11 11:50 tesla_footprinting.xml

┌──(root㉿kali-2026)-[~]
└─#
```

Screenshot 5: JSON results inspection

```
┌──(root㉿kali-2026)-[~]
└─# cat tesla_footprinting.json
{"cmd":"-d tesla.com -b crtsh -f tesla_footprinting","hosts":["*.cn.tesla.com","*.de.tes
la.com","*.eu.logs.tesla.com","*.gf.tesla.com","*.logs.tesla.com","*.na.logs.tesla.com",
"*.nv.tesla.com","*.ny.tesla.com","*.paloalto.tesla.com","*.powerhub.energy.tesla.com","
*.tx.tesla.com","CitiApiPgpProdV8.tesla.com","CitiApiSslProdV8.tesla.com","akamai-apigat
eway-einvoicing-stg.tesla.com","akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com",
"akamai-apigateway-vehicleextinfogw-stgsvc-st.tesla.com","ams13-gpgw1.tesla.com","apac-c
ppm.tesla.com","apac.logs.tesla.com","apf-api.eng.vn.cloud.tesla.com","apf-api.prd.vn.cl
oud.tesla.com","api-firebolt.tesla.com","assets.github-ap.tesla.com","assets.github-fw.t
esla.com","assets.github-it.tesla.com","assets.github.tesla.com","auth-stage.tesla.com",
"auth.prd.usw.vn.cloud.tesla.com","autobidder.powerhub.energy.tesla.com","avatars.github
-ap.tesla.com","avatars.github-fw.tesla.com","avatars.github-it.tesla.com","avatars.gith
ub.tesla.com","bettertime-stage.tesla.com","bettertime.tesla.com","ca.tesla.com","cloud.
tesla.com","cn.tesla.com","codeload.github-ap.tesla.com","codeload.github-fw.tesla.com",
"codeload.github-it.tesla.com","codeload.github.tesla.com","consul.bottlerocket.tesla.co
m","dal11-gpgw1.tesla.com","de.tesla.com","dev-hermes-qd.tesla.com","digitalassets-accou
nts.tesla.com","digitalassets-contents.tesla.com","digitalassets-energy.tesla.com","digi
talassets-secure.tesla.com","digitalassets-shop.tesla.com","digitalassets-stage.tesla.co
m","digitalassets.tesla.com","discovery.bottlerocket.tesla.com","docker.github-ap.tesla.
com","docker.github-fw.tesla.com","docker.github-it.tesla.com","docker.github.tesla.com"
,"em5886.tesla.com","energy.tesla.com","engage.tesla.com","exp-zcp-cloud-ap-mfgsre.tesla
.com","exp-zcp-cloud-eu-itmfg-staging.tesla.com","exp-zcp-cloud-eu-itmfg.tesla.com","exp
-zcp-cloud-na-itmfg-staging.tesla.com","exp-zcp-cloud-na-itmfg.tesla.com","fleet-api.prd
.na.vn.cloud.tesla.com","fleet-api.prd.usw.vn.cloud.tesla.com","fleet-api.prd.usw2.vn.cl
oud.tesla.com","fleet-auth.prd.vn.cloud.tesla.com","fleet-auth.tesla.com","fleetview.ame
rica.fn.tesla.com","fleetview.europe.fn.tesla.com","fleetview.fn.tesla.com","fleetview.p
rd.america.fn.tesla.com","fleetview.prd.eu.fn.tesla.com","fleetview.prd.europe.fn.tesla.
```

Screenshot 6: Footprinting statistics

```
┌──(root㉿kali-2026)-[~]
└─# python3 -c 'import json; d=json.load(open("tesla_footprinting.json")); print("Total
discovered hosts:",len(d.get("hosts",[]))); print("Email addresses:",len(d.get("emails",
[]))); print("Shodan results:",len(d.get("shodan",[])))'
Total discovered hosts: 236
Email addresses: 0
Shodan results: 0

┌──(root㉿kali-2026)-[~]
└─#
```

Disclaimer

This project was completed for educational purposes as part of a cybersecurity training programme and uses only passive OSINT techniques against publicly available data.